

OLYMPIA, WA, USA

WMO: 727920

Lat: 46.973N

Lon: 122.903W

Elev: 188

StdP: 14.60

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24227

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.0	24.6	14.1	11.3	26.8	19.2	14.5	27.3	24.4	48.1	20.9	45.6	1.2	30	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	87.7	65.7	83.8	64.6	80.1	63.3	67.7	84.3	65.8	81.1	64.2	77.7	6.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	82.3	70.9	60.1	78.0	69.4	58.7	74.3	68.2	31.9	84.4	30.5	80.8	29.3	77.9	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	16.3	13.7	13.5	95.0	5.2	4.0	9.8	97.9	6.8	100.2	3.9	102.4	0.1	105.3
			12.8	70.7	5.3	1.9	9.0	72.0	5.9	73.2	3.0	74.3	-0.8	75.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	40.0	40.7	44.4	48.4	54.5	59.2	64.5	64.4	59.3	50.6	43.5	39.0
	DBStd	10.49	6.11	5.41	4.85	5.04	5.42	5.12	4.87	4.25	4.59	4.85	6.04	6.07
	HDD50	1469	313	262	185	89	18	0	0	0	1	52	207	342
	HDD65	5319	776	681	639	500	328	188	68	64	179	447	644	805
	CDD50	1747	2	2	11	40	156	277	449	447	279	69	13	2
	CDD65	122	0	0	0	0	2	15	52	46	7	0	0	0
	CDH74	2549	0	0	1	23	138	344	951	835	255	2	0	0
	CDH80	834	0	0	0	4	29	114	354	273	60	0	0	0
Wind	WSAvg	5.5	5.7	5.6	6.2	6.0	5.5	5.7	5.1	4.8	4.6	5.1	5.8	5.6
Precipitation	PrecAvg	49.4	8.1	5.5	4.8	3.5	2.0	1.6	0.7	1.1	2.0	4.2	7.8	8.1
	PrecMax	66.6	15.1	15.5	10.0	7.8	5.5	3.7	3.0	5.4	7.6	10.7	15.1	14.3
	PrecMin	35.0	0.3	0.2	0.5	1.3	0.2	0.2	0.0	0.0	0.0	0.4	1.4	2.5
	PrecStd	8.0	3.5	3.1	2.0	1.7	1.2	0.9	0.6	1.3	1.6	2.8	3.5	3.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.4	58.3	68.8	77.3	83.9	89.5	94.9	91.4	86.2	71.9	61.6	55.9
		MCWB	53.1	49.3	54.3	58.8	61.3	65.7	69.9	67.1	64.2	57.3	56.8	53.5
	2%	DB	53.1	54.8	61.4	69.4	77.8	82.9	88.3	86.6	80.9	66.8	57.4	52.3
		MCWB	50.2	48.9	51.0	54.6	60.4	63.2	66.3	65.7	63.0	56.3	54.2	50.4
	5%	DB	50.8	52.2	57.0	63.7	72.6	77.3	83.6	82.7	76.3	62.9	55.1	49.9
		MCWB	48.5	47.7	49.2	52.4	58.0	61.3	65.2	65.0	61.7	55.7	52.5	47.9
	10%	DB	48.9	49.8	53.7	59.1	67.3	72.1	79.1	78.7	72.0	59.9	52.8	47.6
		MCWB	46.8	46.2	48.1	49.9	56.0	59.1	63.4	63.5	60.0	54.6	50.4	45.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	53.6	55.9	59.9	64.1	67.4	71.2	69.3	67.3	61.8	58.4	54.1
		MCDB	55.8	55.7	63.8	74.9	78.3	86.4	91.8	85.5	81.8	66.3	60.8	55.5
	2%	WB	51.1	50.7	52.9	56.1	61.5	64.5	68.0	67.4	64.5	59.0	55.1	50.8
		MCDB	52.2	52.9	58.6	66.9	75.3	80.0	85.2	83.6	76.6	63.7	56.9	51.9
	5%	WB	49.0	48.5	50.9	53.5	59.2	62.1	65.9	65.6	62.8	57.0	53.1	48.2
		MCDB	50.4	50.9	55.5	61.2	70.6	75.3	81.7	80.6	73.2	60.8	54.6	49.6
	10%	WB	47.1	46.4	49.0	51.3	56.9	60.0	64.0	64.0	61.0	55.1	50.9	46.0
		MCDB	48.8	49.1	53.2	57.4	66.0	70.8	77.7	77.3	70.0	58.7	52.5	47.5
Mean Daily Temperature Range	5% DB	MDBR	11.7	15.1	17.4	19.5	21.3	21.6	25.5	25.6	23.8	18.1	13.7	11.1
		MCDBR	13.1	19.8	24.8	30.5	33.6	33.6	35.5	34.9	33.5	24.9	15.1	11.3
		MCWBR	9.9	13.8	14.9	16.5	16.7	15.7	15.8	15.8	17.2	15.3	11.2	9.3
	5% WB	MCDBR	10.3	14.4	20.0	24.9	28.1	29.4	32.3	31.5	27.8	18.5	11.9	10.0
		MCWBR	8.8	11.1	13.1	14.1	14.7	14.5	15.1	14.9	14.8	12.6	9.7	8.8
Clear-Sky Solar Irradiance	taub	0.328	0.315	0.332	0.357	0.380	0.366	0.351	0.355	0.342	0.333	0.328	0.331	
	taud	2.473	2.544	2.509	2.409	2.362	2.425	2.487	2.478	2.520	2.544	2.497	2.407	
	Ebn at Noon	241	274	284	285	281	285	288	282	277	263	239	222	
	Edh at Noon	21	24	29	35	37	36	33	32	28	23	21	20	
All-Sky Solar Radiation	RadAvg	328	598	885	1296	1642	1798	2009	1720	1269	723	386	271	
	RadStd	51	82	110	122	140	182	153	123	117	82	51	40	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page